Step 1: Explore Confidentiality

Objective: Understand how confidentiality is maintained and identify security violations when unauthorized access occurs.

1. Create a File Containing Sensitive Information

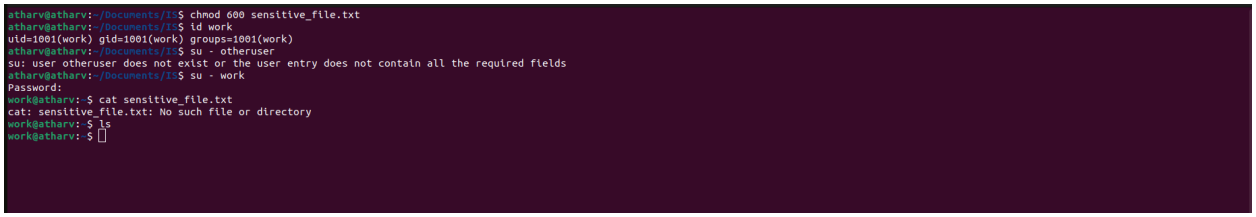
- Open a terminal or command prompt.
- Create a text file named sensitive_file.txt:
  Linux:
  echo "Confidential Data: Usernames and Passwords" > sensitive_file.txt
  Windows:
  Open Notepad, type Confidential Data: Usernames and Passwords, and save the file as sensitive_file.txt.

2. Restrict File Permissions

- Set file permissions so only the owner can access it:
  Linux:
  chmod 600 sensitive_file.txt
  Explanation: The 600 permission allows the owner to read/write the file but denies access to others.
  Windows:
  Right-click the file → Properties → Security → Edit permissions → Deny access for all users except the owner.

3. Simulate Unauthorized Access

- Switch to another user or simulate unauthorized access:
  Linux:
  su - otheruser
  cat sensitive_file.txt
  Windows:
  Switch user accounts or create a new user, then try opening the file.
- Observation:
  Expected error message: "Permission denied"
- Discussion:
  How this protects confidentiality.
  Real-world examples, such as protecting medical or financial records.



Step 2: Analyze Integrity

Objective: Understand how data integrity can be compromised and verify its integrity using
hashing.

1. Create or Access a Log File

   Use an existing log file or create a simulated one:
   Linux: sudo nano /var/log/syslog (requires root access).
   Windows: Open Event Viewer (eventvwr) or create a text file named 19logfile.txt.

2. Modify the Log File (Simulate Unauthorized Changes)

   Add or change log entries to simulate a security violation:
   Linux: Edit the file: sudo nano /var/log/syslog → Add a fake entry:
   Jan 1 12:00:00 UnauthorizedAccess: Admin login.
   Windows: Open logfile.txt in Notepad and add UnauthorizedAccess: Admin login.

3. Verify Integrity with Hashing

   Calculate the file's hash before and after modification:
   Linux: Use sha256sum logfile.txt and note the hash.
   Windows: Use PowerShell: Get-FileHash .\logfile.txt -Algorithm SHA256

Observe the hash difference.
Discussion: Discuss how unauthorized changes compromise data integrity. Relate this to tampering scenarios in real-world applications, such as altering financial records or audit logs.

```
  GNU nano 6.2                                                    /var/log/syslog *
Mar 22 18:00:49 atharv kernel: [    7.277789] rfkill: input handler disabled
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.gtk.vfs.Daemon' requested by ':1.19' (uid=128 pid=1900 comm="ibus-daemo
Mar 22 18:00:49 atharv gnome-shell[1821]: JS ERROR: TypeError: this._managerProxy is undefined#012_onGeoclueVanished@resource:///org/gnome/shell/ui/status/location.js:169:9
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.freedesktop.systemd1' requested by ':1.10' (uid=128 pid=1890 comm="/usr
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activated service 'org.freedesktop.systemd1' failed: Process org.freedesktop.systemd1 exited with st
Mar 22 18:00:49 atharv gsd-sharing[1890]: Failed to StopUnit service: GDBus.Error:org.freedesktop.DBus.Error.Spawn.ChildExited: Process org.freedesktop.systemd1 exited with status 1
Mar 22 18:00:49 atharv gsd-sharing[1890]: Failed to StopUnit service: GDBus.Error:org.freedesktop.DBus.Error.Spawn.ChildExited: Process org.freedesktop.systemd1 exited with status 1
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Successfully activated service 'org.gtk.vfs.Daemon'
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.freedesktop.portal.IBus' requested by ':1.19' (uid=128 pid=1900 comm="i
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Successfully activated service 'org.freedesktop.portal.IBus'
Mar 22 18:00:49 atharv dbus-daemon[737]: [system] Activating via systemd: service name='org.freedesktop.ColorManager' unit='colord.service' requested by ':1.51' (uid=128 pid=1898 comm="/usr/libexec/gsd-c
Mar 22 18:00:49 atharv systemd[1]: Starting Manage, Install and Generate Color Profiles...
Mar 22 18:00:49 atharv spice-vdagent[2000]: vdagent virtio channel /dev/virtio-ports/com.redhat.spice.0 does not exist, exiting
Mar 22 18:00:49 atharv gnome-session-binary[1795]: Entering running state
Mar 22 18:00:49 atharv NetworkManager[738]: <info>  [1742646649.9766] agent-manager: agent[9a8ec2cb97ddaaa9,:1.40/org.gnome.Shell.NetworkAgent/128]: agent registered
Mar 22 18:00:49 atharv gnome-shell[1821]: Error looking up permission: GDBus.Error:org.freedesktop.portal.Error.NotFound: No entry for geolocation
Mar 22 18:00:49 atharv xbrlapi.desktop[2003]: openConnection: connect: No such file or directory
Mar 22 18:00:49 atharv xbrlapi.desktop[2003]: cannot connect to braille devices daemon brltty at :0
Mar 22 18:00:49 atharv dbus-daemon[737]: [system] Successfully activated service 'org.freedesktop.ColorManager'
Mar 22 18:00:49 atharv systemd[1]: Started Manage, Install and Generate Color Profiles.
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0457] device (wlo1): supplicant interface state: scanning -> authenticating
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0457] device (p2p-dev-wlo1): supplicant management interface state: scanning -> authenticating
Mar 22 18:00:50 atharv kernel: [    7.406328] wlo1: send auth to 9c:d4:a6:f4:44:be (try 1/3)
Mar 22 18:00:50 atharv dbus-daemon[737]: [system] Successfully activated service 'org.freedesktop.GeoClue2'
Mar 22 18:00:50 atharv systemd[1]: Started Location Lookup Service.
Mar 22 18:00:50 atharv wpa_supplicant[769]: wlo1: Trying to associate with 9c:d4:a6:f4:44:be (SSID='HUN_5GJio' freq=5200 MHz)
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0487] device (wlo1): supplicant interface state: authenticating -> associating
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0487] device (p2p-dev-wlo1): supplicant management interface state: authenticating -> associating
Mar 22 18:00:50 atharv kernel: [    7.409319] wlo1: authenticated
Mar 22 18:00:50 atharv kernel: [    7.410302] wlo1: associate with 9c:d4:a6:f4:44:be (try 1/3)
Mar 22 18:00:50 atharv kernel: [    7.422422] wlo1: RX AssocResp from 9c:d4:a6:f4:44:be (capab=0x1111 status=0 aid=3)
Mar 22 18:00:50 atharv dbus-daemon[737]: [system] Successfully activated service 'org.freedesktop.UPower'
Mar 22 18:00:50 atharv systemd[1]: Started Daemon for power management.
Mar 22 18:00:50 atharv kernel: [    7.446652] wlo1: associated
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.gnome.ScreenSaver' requested by ':1.26' (uid=128 pid=1930 comm="/usr/lib
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:playback-repeat
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:rotate-video-lock
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:playback-random
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:hibernate
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Successfully activated service 'org.gnome.ScreenSaver'
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): EDID vendor "CMN", prod id 5432
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Printing DDC gathered Modelines:
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0  138.78  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (66.7 kHz eP)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0   92.52  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (44.5 kHz e)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): EDID vendor "CMN", prod id 5432
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Printing DDC gathered Modelines:
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0  138.78  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (66.7 kHz eP)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0   92.52  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (44.5 kHz e)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (--) NVIDIA(GPU-0): DFP-0: disconnected
Jan 1 12:00:00 UnauthorizedAccess: Admin login.
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (--) NVIDIA(GPU-0): DFP-0: Internal TMDS
```

```
atharv@atharv:~/Documents/IS$ sudo nano /var/log/syslog
atharv@atharv:~/Documents/IS$ ls
sensitive_file.txt
atharv@atharv:~/Documents/IS$ sha256sum /var/log/syslog
a126b2b6740a4c8e7b385246ebcdb129a9d5bcde1571f533dfc12b1e71e44c00  /var/log/syslog
atharv@atharv:~/Documents/IS$
```

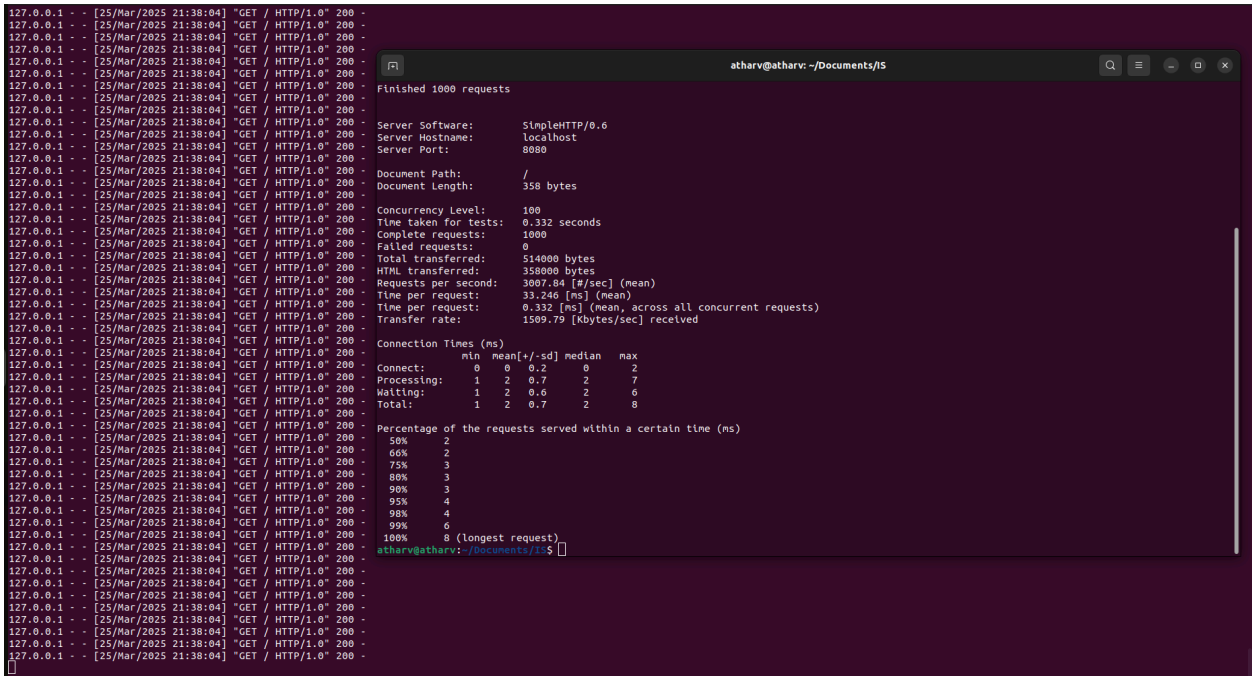
Step 3: Examine Availability
Objective: Explore how system availability is affected during an attack or resource overload.

1. Set Up a Simple Web Server (Optional)
Linux: Use Python to start a basic web server: python3 -m http.server 8080
Windows: Use IIS or WAMP/XAMPP to set up a local server.

2. Simulate Denial-of-Service (DoS) Attack
Use a tool to overload the server with requests:
Linux: Install and use ab (Apache Benchmark):
ab -n 1000 -c 100 http://localhost:8080/
-n: Total number of requests.
-c: Number of concurrent requests.
Windows: Use a custom PowerShell script or any load-testing tool like JMeter.

3. Monitor Server Behavior
Observe the server response time during the attack:
Linux: Check server logs or terminal output for delays or errors.
Windows: Use Task Manager or Resource Monitor to track CPU and network usage.
Note any timeouts or connection refusals.
204. Restore Normal Operations
Stop the attack and ensure normal availability:
Linux: Terminate the ab command or server process (Ctrl+C).
Windows: Stop the server or restart it via IIS Manager.
Verify that the server is responsive again.

5. Discussion
Relate this to real-world examples of denial-of-service attacks on popular websites or applications. Discuss strategies to mitigate such attacks, such as rate-limiting, firewalls, or load balancers.



Step 4: Document Observations

a. For each step, document:

- Actions performed.
- Observations (e.g., error messages, response times).
- Screenshots of commands and results.

b. Analyze how each violation impacts the overall security of the system.